CECS 229 – Assignment 1

Congruence Function

Given the file, 'pa1.py', complete the code for the following function:

1. `congruent_to(a, m, low, high)` – this function returns a list of all the integers that are congruent to a (mod m) that are in the range from low to high, inclusively.
    a. a – is the dividend.
    b. m – is the modulus.
    c. low – the lower bound.
    d. high – the upper bound.
    e. The list returned should be the numbers that are congruent to a under mod m that lie inclusively within the bounds of low to high.

All integers, x, that are congruent to a under modulo m must satisfy:

$$x - a = m * k, \text{ for some integer k.}$$

This means that all congruent values must be:

$$low \leq m * k + a \leq high$$

Example Input/Output:

```
congruent_to(3, 5, -10, 15)
[-7, -2, 3, 8, 13]
congruent_to(12, 15, -29, -11)
[-18]
congruent_to(-20, 4, 3, 21)
[4, 8, 12, 16, 20]
```

Notes:

1. Please place your name in the comment block at the top of the provided file.
2. Fill in the necessary code to complete the functions. You should not modify any of the provided code other than in the specified locations (ie. the TODOs).
3. Your solutions may not use any built-in functions that perform the equivalent task.
4. Test your code using the provided 'main.py' file. You can write additional tests in the 'your_tester.py' file to further check your solutions. Note: just because the main tests pass does not guarantee that it will pass the tests on Gradescope.
5. When you are finished, submit your 'pa1.py' file to Gradescope on Canvas for further testing and grading. It should allow you to make repeated submissions.